

3.6 Directional Derivatives and Gradients Worksheet

1. True or False: The directional derivative of $f(x, y)$ at the point (x_1, y_1) in the direction of θ_1 is given by $f_x(x_1, y_1)\sin(\theta_1) + f_y(x_1, y_1)\cos(\theta_1)$.

FALSE. We know that the directional derivative of $f(x, y)$ at the point (x_1, y_1) in the direction of θ_1 is given by $f_x(x_1, y_1)a + f_y(x_1, y_1)b$, where $\mathbf{u} = \langle a, b \rangle$ a unit vector pointing in the direction of θ_1 . This is equivalent to the expression $f_x(x_1, y_1)\cos(\theta_1) + f_y(x_1, y_1)\sin(\theta_1)$.

2. Find the directional derivative of $f(x, y, z) = 3x^2y^2 + 2yz$ at the point $(-1, 0, 4)$ pointing in the direction $\langle -1, 3, 3 \rangle$.

We note that a unit vector in the direction of $\langle -1, 3, 3 \rangle$ is $\mathbf{u} = \frac{1}{\|\langle -1, 3, 3 \rangle\|} \langle -1, 3, 3 \rangle = \left\langle \frac{-1}{\sqrt{19}}, \frac{3}{\sqrt{19}}, \frac{3}{\sqrt{19}} \right\rangle$.

We also note that the gradient of f is $\vec{\nabla} f(x, y, z) = \langle 6xy^2, 6x^2y + 2z, 2y \rangle$.

So, $D_{\mathbf{u}}f(-1, 0, 4) = \vec{\nabla} f(-1, 0, 4) \cdot \mathbf{u} = \langle 0, 8, 0 \rangle \cdot \left\langle \frac{-1}{\sqrt{19}}, \frac{3}{\sqrt{19}}, \frac{3}{\sqrt{19}} \right\rangle = \frac{24}{\sqrt{19}}$.

3. Let $f(x, y) = 20 - (2x^2 + y^2)$.

- a) Find a vector at the point $(1, 3)$ pointing in the direction in which f increasing fastest?

We know that the gradient will be a vector pointing in the direction of greatest increase. So we compute

$$\nabla f = \langle -4x, -2y \rangle \text{ and therefore } \nabla f(1, 3) = \langle -4, -6 \rangle.$$

- b) What is the maximum rate of change?

The maximum rate of change will be the magnitude of the gradient and so we calculate

$$\text{max rate of change} = \|\nabla f(1, 3)\| = \|\langle -4, -6 \rangle\| = \sqrt{16 + 36} = \sqrt{52}$$

4. Let $f(x, y) = xye^{x^3+x+y^2}$.

a) Find ∇f .

We see that:

$$\begin{aligned}\nabla f &= \left\langle ye^{x^3+x+y^2} + xye^{x^3+x+y^2}(3x^2+1), xe^{x^3+x+y^2} + xye^{x^3+x+y^2}(2y) \right\rangle \\ &= \left\langle ye^{x^3+x+y^2}[1+3x^2+x], xe^{x^3+x+y^2}[1+2y^2] \right\rangle\end{aligned}$$

b) Find a vector that is orthogonal to the curve $f(x, y) = e^3$ at the point $(1, 1)$.

We know that the gradient creates a vector which is orthogonal to the surface $f(x, y)$ at any point.

Therefore, since the curve $f(x, y) = e^3$ is just a level curve of $f(x, y)$ we know that $\nabla f(1, 1)$ will still be a vector that is orthogonal to the curve! Therefore, the vector $\nabla f(1, 1) = \langle 5e^3, 3e^3 \rangle$ is orthogonal to the curve at $(1, 1)$.

5. Let $\mathbf{u} = \langle 1, 2, -1 \rangle$. Let $f(x, y, z) = \tan^{-1}(xy) + e^{z^2+x}$. Find the derivative of f in the direction of $\mathbf{u}$ at the point $(1, 1, 1)$.

We note that a unit vector in the direction of $\mathbf{u}$ is $\mathbf{v} = \frac{1}{\|\mathbf{u}\|} \mathbf{u} = \left\langle \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}} \right\rangle$.

We also note that the gradient of f is $\bar{\nabla} f(x, y, z) = \left\langle \frac{y}{1+x^2y^2} + e^{z^2+x}, \frac{x}{1+x^2y^2}, 2ze^{z^2+x} \right\rangle$.

So,

$$\begin{aligned}D_{\mathbf{v}} f(1, 1, 1) &= \bar{\nabla} f(1, 1, 1) \cdot \mathbf{v} = \left\langle \frac{1}{2} + e^2, \frac{1}{2}, 2e^2 \right\rangle \cdot \left\langle \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}} \right\rangle \\ &= \left(\frac{1}{2} + e^2 \right) \frac{1}{\sqrt{6}} + \frac{1}{\sqrt{6}} - \frac{2e^2}{\sqrt{6}} \\ &= \frac{1}{\sqrt{6}} \left[\frac{1}{2} + e^2 + 1 - 2e^2 \right] \\ &= \frac{1}{\sqrt{6}} \left[\frac{3}{2} - e^2 \right]\end{aligned}$$

6. Find a unit vector which is normal to the surface $z = x^2y^2 + y + 1$ at the point $(0,0,1)$.

We know that the gradient creates a vector which is normal to the surface at any point. To find the gradient we define a function $f(x, y, z) = z - (x^2y^2 + y + 1) = z - x^2y^2 - y - 1$. Then we calculate that

$$\nabla f = \langle -2xy^2, -2yx^2 - 1, 1 \rangle.$$

So $\nabla f(0,0,1) = \langle 0, -1, 1 \rangle$ is a vector that is normal to the surface at the point $(0,0,1)$.

To make this a unit vector we compute $\mathbf{u} = \frac{\nabla f(0,0,1)}{\|\nabla f(0,0,1)\|} = \frac{\langle 0, -1, 1 \rangle}{\sqrt{2}} = \left\langle 0, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$.

7. Let $f(x, y, z) = \frac{x}{y} - yz$. Find the directions in which $f(x, y, z)$ increases and decreases most rapidly at the point $(4,1,1)$, and then find the directional derivatives of $f(x, y, z)$ along these directions.

$$\text{We compute } \nabla f = \left\langle \frac{1}{y}, \frac{-x}{y^2} - z, -y \right\rangle.$$

We know that the gradient represents the direction in which $f(x, y, z)$ increases most rapidly and $-\nabla f$ is the direction in which $f(x, y, z)$ decreases most rapidly.

Therefore, at the point $(4,1,1)$ we have: $\nabla f(4,1,1) = \langle 1, -5, -1 \rangle$ and $-\nabla f = \langle -1, 5, 1 \rangle$.

To compute the directional derivatives, we will need to think about unit vectors pointing in these directions. Let $\mathbf{u}$ represent the unit vector in the direction of $\nabla f(4,1,1)$.

Let $\mathbf{v}$ represent the unit vector in the direction of $-\nabla f(4,1,1)$.

Then, by the last theorem in section 3.6, we know that:

$$D_{\mathbf{u}}f(4,1,1) = \|\nabla f(4,1,1)\| = \sqrt{27} \text{ and}$$

$$D_{\mathbf{v}}f(4,1,1) = -\|\nabla f(4,1,1)\| = -\sqrt{27}$$

8. Let $f(x, y, z) = 2ye^{x-2y} + \ln(z)$

- a) Find the directional derivative of $f(x, y, z)$ at the point $(2, 1, 1)$ along the direction of the vector $\langle -3, 1, 2 \rangle$.

We must first convert $\langle -3, 1, 2 \rangle$ to a unit vector. We calculate $\mathbf{u} = \left\langle \frac{-3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}} \right\rangle$.

We also calculate that $\nabla f = \left\langle 2ye^{x-2y}, 2e^{x-2y} + 2ye^{x-2y}(-2), \frac{1}{z} \right\rangle = \left\langle 2ye^{x-2y}, 2e^{x-2y}(1-2y), \frac{1}{z} \right\rangle$

Therefore, $D_{\mathbf{u}}f(2, 1, 1) = \nabla f(2, 1, 1) \cdot \mathbf{u} = \langle 2, -2, 1 \rangle \cdot \left\langle \frac{-3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}} \right\rangle = \frac{-6}{\sqrt{14}} - \frac{2}{\sqrt{14}} + \frac{2}{\sqrt{14}} = -\frac{6}{\sqrt{14}}$

- b) Is the direction derivative in part(a) maximum among all directional derivatives of $f(x, y, z)$ at $(2, 1, 1)$? If not, find the maximum one and the direction along which it appears.

The maximum directional derivative is always in the direction of the gradient.

Therefore, the maximum directional derivative at the point is given by $\|\nabla f(2, 1, 1)\| = \sqrt{4 + 4 + 1} = 3$.

So, we can conclude that the answer from part (a) is *not* the maximum among all directional derivatives at $(2, 1, 1)$.

The direction in which the maximum occurs is given by the unit vector $\frac{\nabla f(2, 1, 1)}{\|\nabla f(2, 1, 1)\|} = \left\langle \frac{2}{3}, \frac{-2}{3}, \frac{1}{3} \right\rangle$.

- c) Find a direction $\mathbf{v}$ such that $D_{\mathbf{v}}f(2, 1, 1) = 0$.

We want to find a vector $\mathbf{v}$ such that $\nabla f(2, 1, 1) \cdot \mathbf{v} = 0 \rightarrow \langle 2, -2, 1 \rangle \cdot \mathbf{v} = 0$. Many different vectors for $\mathbf{v}$ are possible. One possible selection of $\mathbf{v}$ would be $\mathbf{v} = \langle 1, 1, 0 \rangle$.

9. The temperature in degrees Celsius on the surface of a metal plate is $T(x, y) = 20 - 4x^2 - y^2$ where x and y are measured in centimeters. In what direction from the point $(2, -3)$ does the temperature decrease most rapidly? What is this rate of decrease?

The direction of the greatest decrease would be in the direction of $-\nabla T$.

We calculate that $\nabla T = \langle -8x, -2y \rangle$.

Therefore, the direction in which the temperature decreases most rapidly is $-\nabla T(2, -3) = \langle 16, -6 \rangle$.

The rate of this decrease is given by $-\|\nabla T(2, -3)\| = -\sqrt{16^2 + 6^2} = -\sqrt{292}$